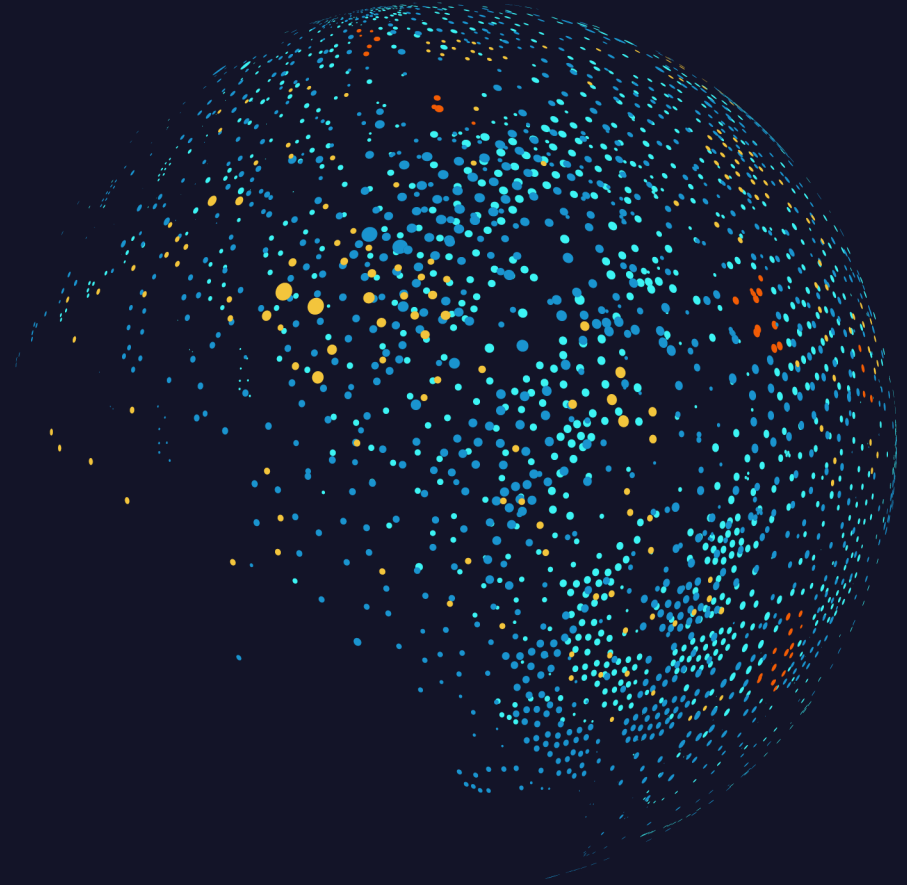


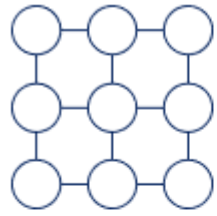
Epidemiological data harmonisation: Global Repository of Epidemiological Parameters



**Climate-Sensitive Vector Dynamics Modelling
Workshop**
WHO Hub for Pandemic and Epidemic Intelligence
18 September 2025

Collaboratory vision

A world where interconnected pandemic and epidemic intelligence communities collaboratively and rapidly respond with enhanced data, better analysis, and actionable insights.



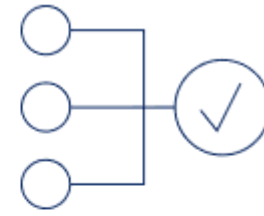
Analysis with
better data

Trusted network to share
data/results and information,
including context data



Analysis
improved

Joint, transparent, reusable
and comparable advanced federated
analysis supported



Analysis used
for decisions

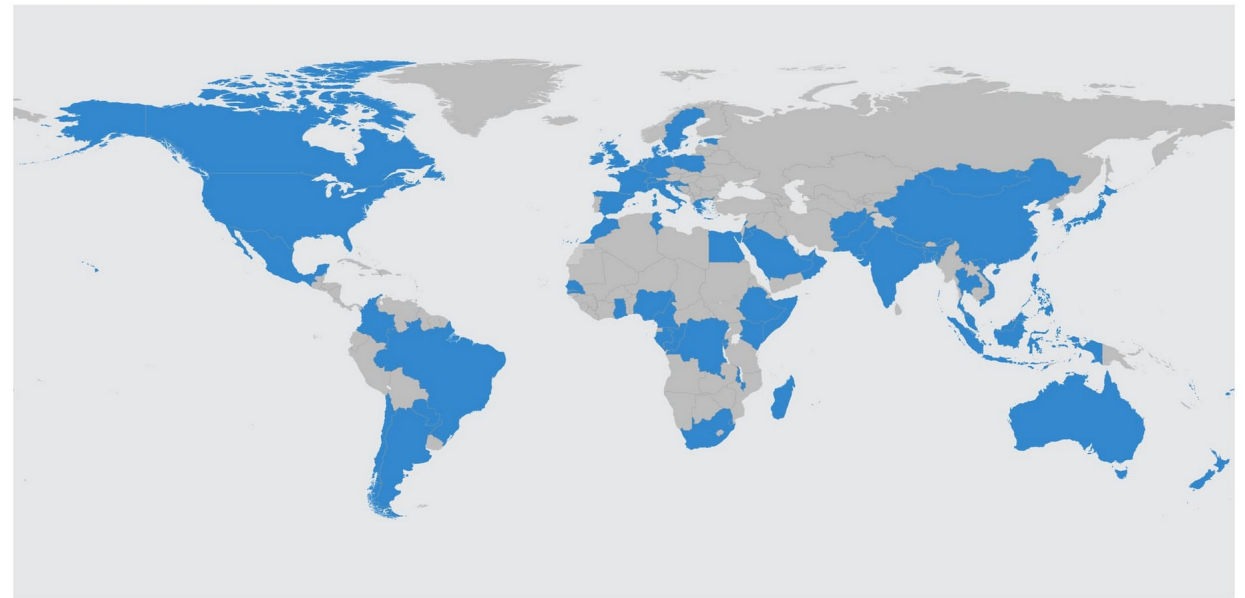
Collaboratory
Pandemic and Epidemic Intelligence
EpiParameter Community

Collaboratory communities and members

Since launching the Collaboratory platform in March 2024 we have **15** active Communities of Practice and **1200+** Collaboratory members.

Members include modellers, epidemiologists, data scientists, developers, public health experts, and decision makers from **100+** ministries of health (public health agencies), international organisations and academic institutions from more than **60** countries and territories.

Collaboratory Membership



The designations employed and the presentation of the material in this publication do not imply the expression of any opinion whatsoever on the part of WHO concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted and dashed lines on maps represent approximate border lines for which there may not yet be full agreement.

Data Source: World Health Organization
Map Production: WHO Health Emergencies Programme
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Epi Parameter Community of Practice

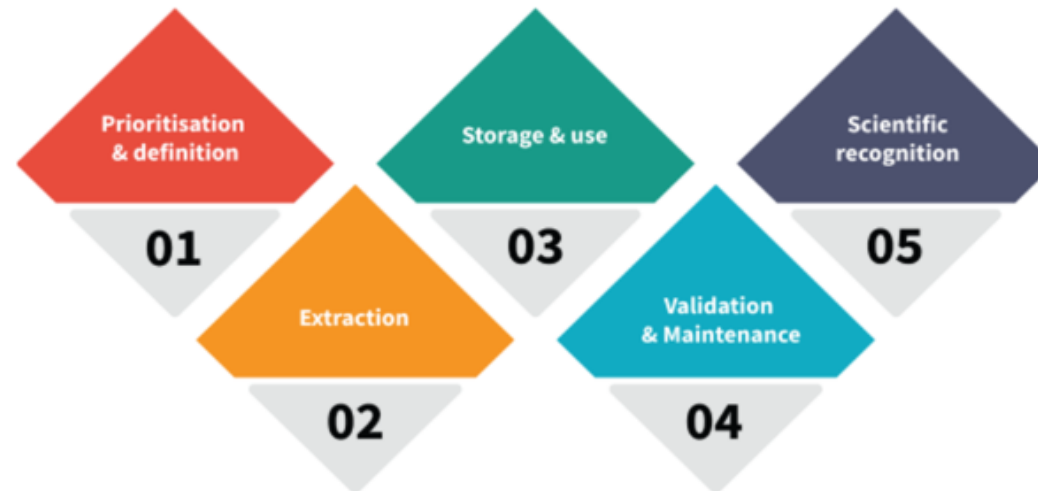
Problem Statement

There is currently no global repository of epidemiological parameters that the global modelling community can access, use, and contribute to.



Future State

A global repository of epidemiological parameters that is publicly accessible by modellers, epidemiologist, subject matter experts and decision makers to inform public health response.



Why grEPI for VBDs?

Data harmonisation and decision-support tools

Collecting data on vectors, hosts and human disease is vital for VBD control, while the availability and accessibility of these data to researchers and decision-makers is important for quantifying risks and designing effective solutions. However, data needs to be harmonised for effective use in infectious disease decision-support modelling tools.⁵⁷ Harmonisation allows for seamless linking of health data to climatic and environmental datasets, along with key socio-economic risk factors at common space-time scales.

Strengthening resilience to emerging vector-borne diseases in Europe: lessons learnt from countries facing endemic transmission

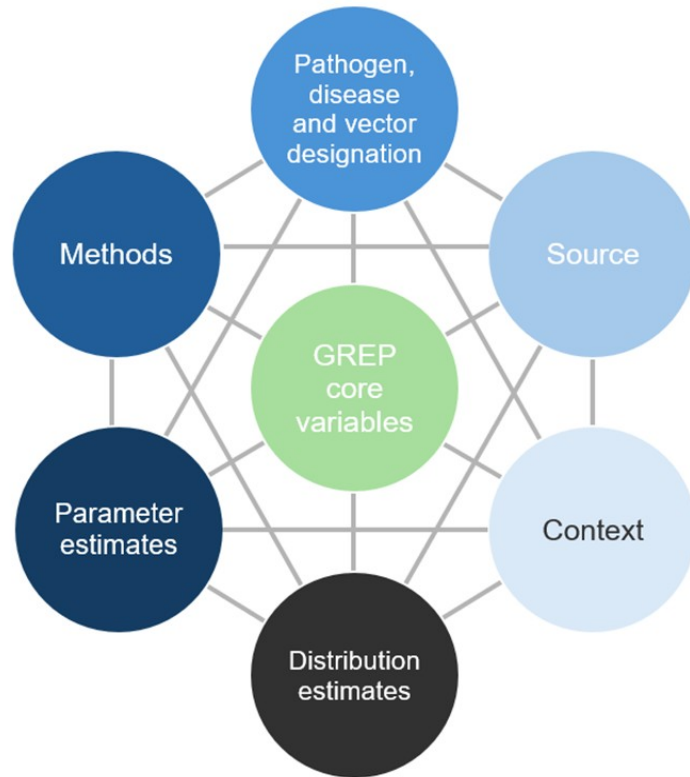


Gina E. C. Charnley,^{a,b,*} Tilly Alcayna,^{a,c,d} Alex Almuedo-Riera,^{a,f} Christiana Antoniou,^g Athanase Badolo,^h Frederic Bartumeus,^{i,j} Laura-Lee Boodram,^k Rubén Bueno-Marí,^{l,m} Claudia Codeço,ⁿ Flavio Codeço Coelho,^o Federico Costa,^p Horace Cox,^k Nabil Haddad,^q Nurulhusna Ab Hamid,^r Pattamaporn Kittayapong,^s Gülay Korukluoğlu,^t Antonios Michaelakis,^u Rafael Maciel-de-Freitas,^{v,w} Tomas Montalvo,^{x,y} Jose Muñoz,^z Silvia Saucedo Oliveras,^z John R. B. Palmer,^{aa} Carlos Jesús Barboza Pizard,^{ab} Guilherme S. Ribeiro,^{ac,ad} and Rachel Lowe^{a,c,j,**}



What is in scope for grEPI?

Core variables



The Global Repository of EPIdemiological parameters (grEPI) will contain key epidemiological parameters, for example:

- serial interval,
- generation time,
- attack rate,
- reproduction number,
- ...and more.

Parameters will be able to be compared and contrasted across a range of settings (community, hospital, other) and countries.

Pathogens

Imperial College London Pathogen Epidemiology Review Group Priority Pathogens Project

- **Completed**
 - Marburg
 - Lassa Mammarenavirus
 - SARS-CoV-1
 - Ebola
- **Ongoing**
 - Zika
 - MERS-CoV
 - Rift Valley Fever
 - Nipah

Collaboratory Communities of Practice

- **Ongoing**
 - Mpox (August 2024)
 - Dengue (July 2025)
 - SARS-CoV-2 (upcoming)
 - H5N1 (upcoming)

Public Health Agency of Canada

- **Ongoing**
 - Measles

REVIEW · Volume 24, Issue 12, E762-E773, December 2024

[Download Full Issue](#)

Ebola virus disease mathematical models and epidemiological parameters: a systematic review

[Rebecca K Nash, MSc^{a,*}](#) · [Sangeeta Bhatia, PhD^{a,b,c,*}](#) · [Christian Morgenstern, MSci^a](#) · [Patrick Doohan, PhD^a](#) · [David Jorgensen, PhD^a](#) · [Kelly McCain, MSPH^a](#) · et al. [Show more](#)

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Parameter types



In 2026 we will be designing the database structure for temperature dependent parameters – we welcome your contributions!

Global Repository of Epidemiological Parameters

Epidemiological Parameters

Reference Data

Home / Reference Data

Search...

Epidemiological Parameter estimate type and subtype

Epidemiological Parameter estimate unit

Epidemiological Parameter estimate value type

Epidemiological Parameter estimate uncertainty unit

Epidemiological Parameter estimate uncertainty value type

Epidemiological Parameter estimate distribution type

Epidemiological Parameter Method Reproduction number

Epidemiological Parameter estimate method CFR IFR

Epidemiological Parameter estimate method Moment value

Disease - modes of transmission

Post V1 release

Temperature value (or range)

Post V1 release

Temperature value units (e.g. °C)

Parameter types

Attack rate

- Primary
- Secondary

Doubling time

Growth rate

Halving time

Human delay

- Generation time
- Incubation period
- Infectious period
- Latent period
- Serial interval
- Other

Mutations

- Evolutionary rate
- Mutation rate
- Substitution rate

Dispersion

- Overdispersion

Relative contribution

Reproduction number

- Basic R_0
- Effective R_e

Severity

- Case fatality rate (CFR)
- Infection fatality rate (IFR)

Seroprevalence

- Immunofluorescence assay (IFA)
- Immunoglobulin G (IgG)
- Immunoglobulin M (IgM)
- Healthcare-associated infections
- Plague reduction neutralization tests
- Unspecified



Vector

- Extrinsic incubation period

Parameter types

1 AedesTraits: A global database of temperature–dependent trait
2 responses in *Aedes* mosquitoes

3 Daniele Da Re^{1,*}✉, Veronica Andreo^{2,3,*}, Tomas San Miguel^{2,3}, Marharyta Blaha⁴, Roberto
4 Rosà⁴, Annapaola Rizzoli¹, Joe Harrison⁵, Sean Sorek⁵, Leah R. Johnson⁵, and Paul J.
5 Huxley^{5,6}✉

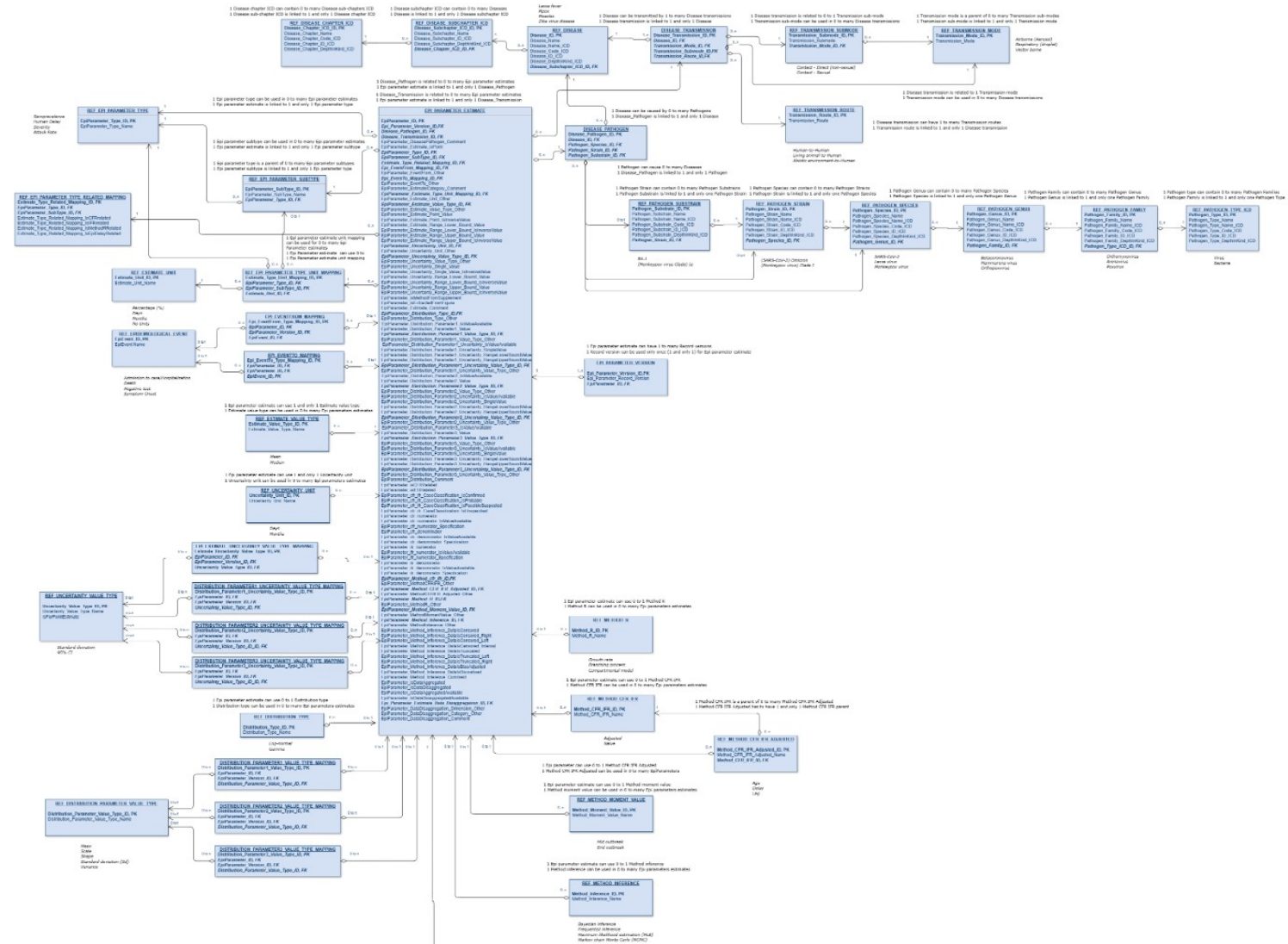
For us to consider:

- Relational elements of these parameters
- Interface with human parameters & how to link these in a meaningful way

Trait Category	Trait Examples
Morphology	- Body size (mean dry weight, wing length, wing size, wet mass, cephalothorax length, head width, live body weight, body length, wing area) - Egg size (non-diapause/diapause egg length and width under different conditions)
Life History	- Development time (mean, median, min, max duration, hatch to pupation, under stressors like temperature, food quantity, competition) - Fecundity (mean lifetime eggs, eggs per cycle, individual-level lifetime eggs, ovariole count, eggs post blood meal) - Gonotrophic cycle length (duration, number of cycles) - Longevity (mean, max, min lifespan, days when X% population alive) - Survival (percent surviving life stage, egg hatching, pupation rate, survival to adulthood, survival under stressors, mean survival probability) - Blood feeding frequency (mean number of blood meals, percent taking one/two meals) - Pre-bloodmeal period, pre-oviposition period, incubation period, juvenile life span
Physiology	- Sex ratio (as function of temperature, larval food type) - Temperature tolerance (percent survival after extreme temperatures, knock-down time, chill coma onset/recovery, chill injury) - Energy reserves (glucose, glycogen, trehalose content) - Hemolymph composition (Na⁺, K⁺ concentrations after cold stress) - Weight loss during exhaustive flight - Critical photoperiod period (minimum light exposure for egg hatching)
Behaviour	- Biting rate (time to first bite, number of bites, time between bites) - Feeding preference (preference for protein-rich vs. sucrose meals) - Mating capacity (mean % females inseminated by one male)
Infection & Transmission	- Infection rate (percent infected after exposure to DENV, ZIKV, CHIKV, Wolbachia, body/head/salivary gland/midgut presence) - Dissemination rate (percent dissemination to body parts like legs, heads, salivary glands) - Transmission rate & efficiency (vertical/horizontal transmission, progeny infection, transmission to hosts) - Viral titers & viral load (log₁₀ plaque forming units, virus concentration in body, legs, salivary glands) - Extrinsic incubation period & rate (time until transmission potential) - Viral replication rate (as a function of temperature/post-infection days)

Database schema

Under development



Sample extraction process

RESEARCH ARTICLE

Impact of temperature on the extrinsic incubation period of Zika virus in *Aedes aegypti*

Olivia C. Winokur^{1,2}, Bradley J. Main¹, Jay Nicholson^{1a}, Christopher M. Barker^{1*}

Median time from ZIKV ingestion to transmission (Median EIP)

Temperature (°C)	Days
30	5.1
26	9.6
21	24.2
18	N/A

Outbreak strain: Asian lineage strain of ZIKV from Puerto Rico

Comment: EIP at 18 degrees could not be estimated. The majority of ZIKV vector competence work in *Aedes aegypti* has been conducted at temperatures between 26 to 28 degrees Celsius.

Not currently accounted for in grEPI : Ability to note (outside the above comment section) that humidity was held at 70 – 80%

Summary

-  **1 Speed up** modelling/parameter summary processes by accessing readily available, robust parameters from a single source.
-  **2 Enhance model contestability** by challenging model assumptions using parameters from different countries, settings and contexts.
-  **3 Foster the expansion of professional networks** by engaging with the CoP and the various activities that are run within it.

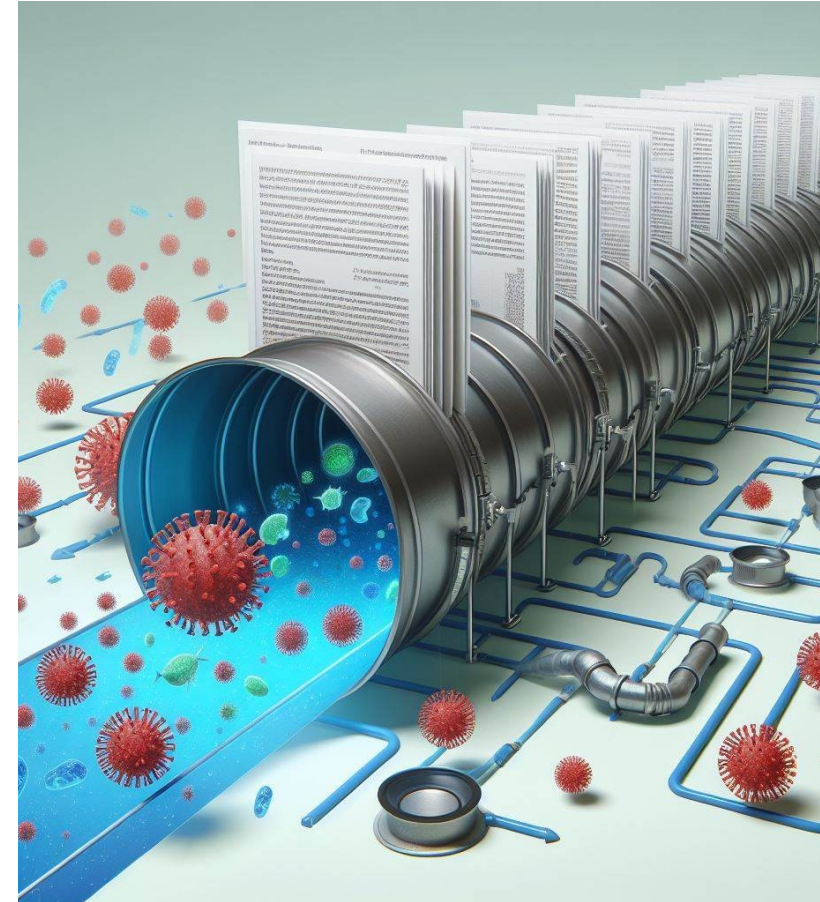
Next steps for grEPI

AI supported parameter extraction

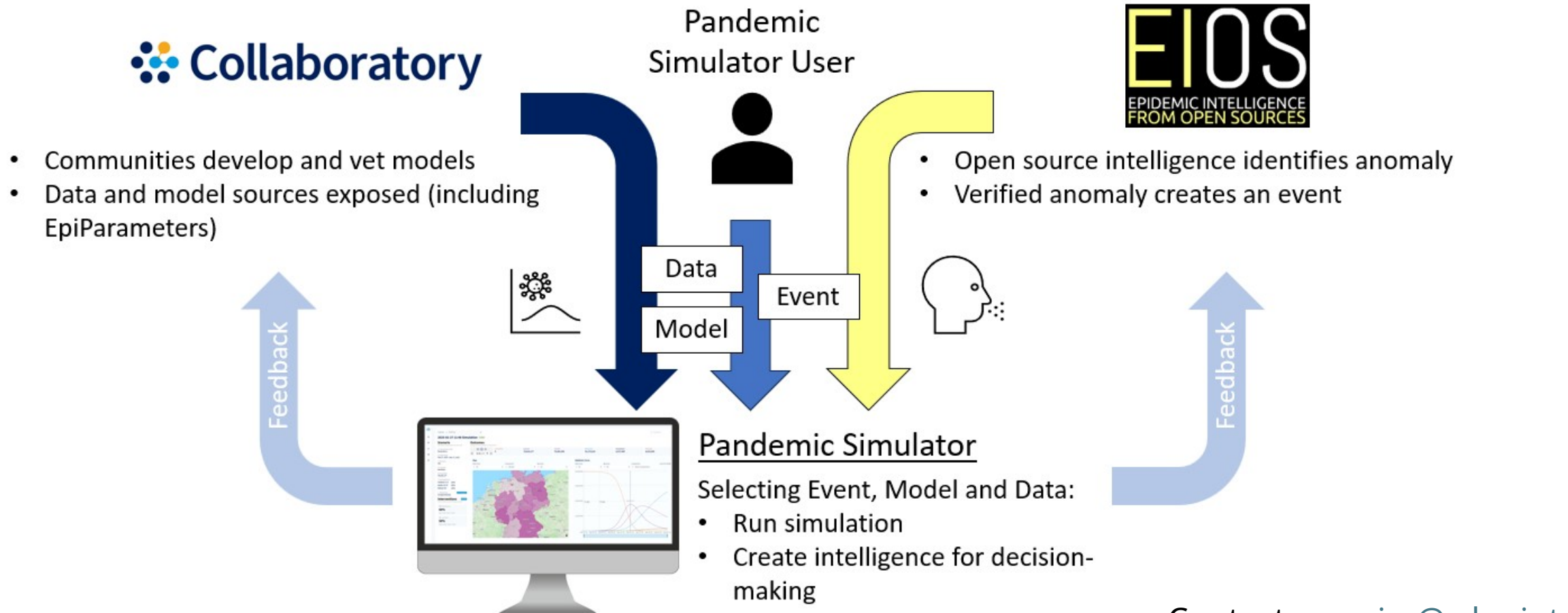
Collaboration with the Public Health Agency of Canada

Current status:

- Reviewers can scroll through a PDF paper alongside the LLM and provide feedback/regenerate responses to improve accuracy
- Multiple LLMs are deployed to extract parameters – when there is a disagreement, human reviewers can provide feedback to finetune the responses
- For now, we still need continuous human review to ensure accuracy and reliability of the extracted data
- We seem to have hit a roadblock for processing old PDFs and are considering using more expensive tools with better OCR



Integration with the Decision Support Simulator



Contact: pansim@who.int

Integration with the Decision Support Simulator



Disease / transmission mode	COVID-19 / Respiratory	Dengue / Vector-borne
Model type	Compartmental, agent-based	Compartmental incl. stochasticity
Interventions	Mask wearing, lockdown, vaccination, Point-of-Interest closure Vector control, bite reduction Can be activated / terminated by date or threshold	

Data sources:

Boundaries: WHO (national), GADM (sub-national)

Population: Worldpop

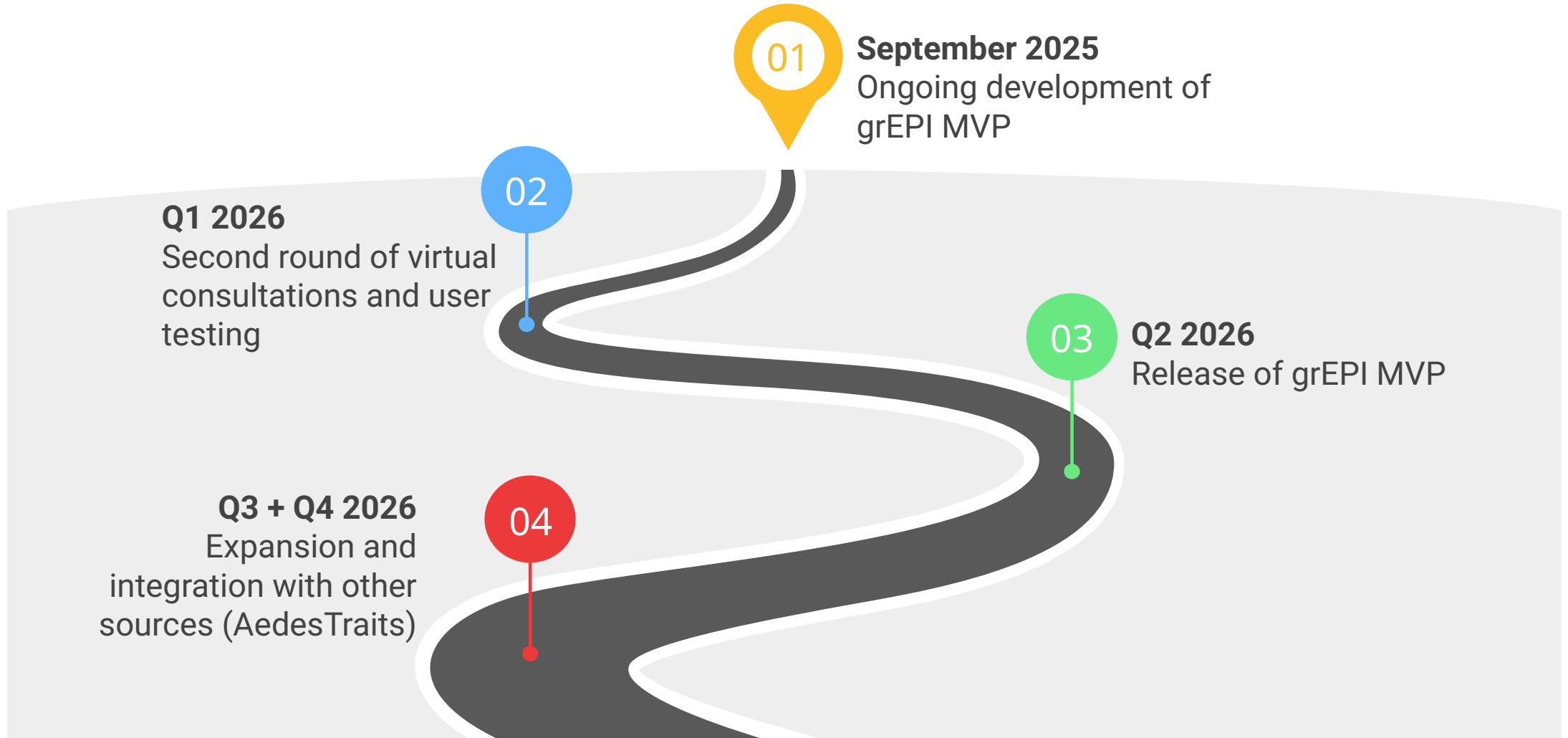
Household composition: GLOPOP-S

POI: OpenStreetMap

Dengue Seroprevalence: ArboMap

Contact: pansim@who.int

grEPI timeline



Upcoming events

Workshops	29 October			
	Testing the underlying Dengue models in the Pandemic Decision Support Simulator			
	Virtual, hosted by WHO			
Webinars	7 October	22 October	13 November	26 November
	Spatial Modelling of Chikungunya	Compound and cascading effects of climatic extremes on dengue outbreak risk	TIMO Citizen Science Study	Gap filling and disaggregation model
	Guilherme Ribeiro, Fiocruz	Chloe Fletcher, BSC	Victoria Ng, PHAC	Ahyoung Lim, LSHTM

Opportunity to collaborate: Joint publication on vector parameter reporting guidelines

Example case study

Mpox product

grEPI parameters available for Mpox analysis

Parameter label	Definition	Example options (based on what is currently in the repository)
PATHOGEN_FK	Selected parameter of interest	Mpox
PARAM_TYPE_FK	Parameter type	Human delay, Reproduction number (Basic R0)
PARAM_SUBTYPE_FK	Parameter sub-type	Incubation period, generation time, All Reproduction number (Basic R0), serial interval
CLADE	Disease clade	I, IIa, IIb (other clades can be added with relevant studies)
PARAM_UNIT_FK	Parameter units	Days, No units
PARAM_VAL	Value of the selected parameter	
PARAM_BOUND_LOWER (PARAM_BOUND_UPPER)	Lower/upper bound of the parameter estimate	
PARAM_VALUE TYPE	Value category	Mean, Median
POP_SEX_FK	Gender filter	Male, Both
POP_AGE_MIN (POP_AGE_MAX)	Minimum/Maximum age of the study population	Min: 20, 23, 28; Max: 61, 64, 71
POP_COUNTRY	Country the estimate was produced from	Netherlands, United States, Democratic Republic of Congo, United Kingdom
ARTICLE_LABEL	First author and year of the study	
ARTICLE_TITLE	Title of the study	

Accessing grEPI in R

```
library(httr)      library(tidyverse)
library(jsonlite)  library(pander)
library(rlist)
```

The core table can then be accessed via the GET() function via the public xmart-api-public-uat.who.int portal:

```
##--Load tables--##
```

```
response1 <- GET("https://xmart-api-public-uat.who.int/COLLAB/MPOX_PARAMETERS")
```

```
#FACT_PARAMETERS
```

```
json1 <- content(response1, "text")
```

```
FACT_PARAMETERS <- data.frame(fromJSON(json1))
```

```
#Tidy column names
```

```
FACT_PARAMETERS <- FACT_PARAMETERS %>% rename_with(str_sub, start = 7L)
```

Obtain a copy of this guidance →



Accessing grEPI in R

```
Ex <- subset(FACT_PARAMETERS, PATHOGEN_FK == 'Mpox' &
  PARAM_TYPE_FK == 'Human delay' &
  PARAM_SUBTYPE_FK == 'incubation period' &
  PARAM_UNIT_FK == 'Days')
```

```
Result <- data.frame(Ex$ARTICLE_LABEL, Ex$PARAM_VAL, Ex$PARAM_BOUND_LOWER,
  Ex$PARAM_BOUND_UPPER, Ex$PARAM_VALUE_TYPE_FK, Ex$POP_COUNTRY)
```

```
colnames(Result) <- c("Article label", "Parameter value", "Lower bound", "Upper
bound", "Value type", "Country")
```

```
Result
## Article label | Parameter value | Lower bound | Upper bound | Value type |
Country
## Miura 2022 | 8.50 | 6.60 | 10.90 | Mean |
Netherlands
## Charniga, 2022 | 7.60 | 6.20 | 9.70 | Mean | United
States
```

...

Obtain a copy of this guidance →



Thank you!

How to participate

Come and chat to me or email us at Collaboratory@who.int to receive a link to join the platform.

Link to more information about the
Epi Parameter initiative

